



Nursing Practice Guideline

Wound Drain Management

Reference #: NPG65

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	Nursing

Introduction

This guideline outlines the procedures and responsibilities of individuals and the health service organisation in relation Wound Drain Management. It ensures the delivery of a uniform, integrated multidisciplinary approach with management that is consistent with best practice.

Risk Statement

Non-compliance with this guideline will:

Breach legislative requirements	☒	Impact on Patient Quality of Care	☒
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☐
Breach SCGH Mission & Values	☒	Staff Safety	☒
Other:	☐		

Table of Contents

Introduction	1
Risk Statement	1
1.0 Principles	1
2.0 Insertion Site Dressing.....	2
3.0 Shortening of Drain.....	2
4.0 Emptying Drain	3
5.0 Removal of Drain	4
6.0 Management of Abdominal Pigtail Drain	5
7.0 Suction Drainage	9
Authorisation & Version Control.....	13

1.0 Principles

Observe the five moments for hand hygiene. Follow aseptic technique and use the risk assessment outlined in the SCGOPHCG Infection Prevention and Management Policy, when handling wound management drains and drain systems. This must include use of appropriate Personal Protective Equipment (PPE) to prevent contamination and mucosal or conjunctival splash injuries.¹⁻⁶

2.0 Insertion Site Dressing

2.1 Safety Information

If the drain is attached to a drainage bag ensure the drainage bag is below the insertion site, DO NOT secure the bag to a bed, use a bedside bag holder to prevent contamination. The drainage bag must not be placed on the floor.

2.2 Nursing Management

- When removing the drain dressing, ensure the drain is anchored to prevent pulling / dragging.⁶
- Drain dressings are to be attended to daily or as per Wound Assessment & Care Plan. Assess for signs or symptoms of complications, e.g., discomfort, loss of skin integrity, odour, haematoma, blockage, signs of infection or dislodgement and document in the inpatient notes and report to a medical officer.¹⁻⁶

2.3 Procedural Information

Requirements

- Wet strength bag
- Dressing pack
- Sterile normal saline 0.9%
- Gauze
- Drain sponge/keyhole dressing
- Adhesive retentive dressing (e.g., Fixomul™)
- Personal Protective Equipment (PPE)

Procedure

1. Perform hand hygiene.
2. Clean the trolley with combined detergent disinfectant wipe. Allow to dry.
3. Gather equipment for the procedure.
4. Perform hand hygiene and prepare the aseptic field.
5. Open dressing pack using corners. Open the remaining equipment onto the aseptic field using a non-touch technique.
6. Don PPE and remove the dressing, assessing amount and type of exudate.
7. Remove PPE and perform hand hygiene.
8. Using forceps, cleanse insertion site with normal saline 0.9%.¹⁻⁶
9. Place drain sponge/keyhole dressing around the insertion site.
10. Cover with gauze if applicable, secure with adhesive dressing and ensure the drain is patent, freely draining and secured.²
11. Dispose of waste.
12. Perform hand hygiene.
13. Clean the trolley with combined detergent disinfectant wipe. Allow to dry.
14. Perform hand hygiene.

3.0 Shortening of Drain

3.1 Safety Information

Identify the type of drain insitu as this will influence whether the drain may be shortened.

To prevent the drain retracting into the wound, ***always insert a sterile safety pin*** into the proximal part of the drain after withdrawal and before cutting off the drain excess.³

Drain requires medical review daily, and the length of shortening is to be documented. Ensure medical staff requesting the drain shortening identify type of drain, and that the drain can be shortened.

If the volume of the drainage is more than the previous day's drainage, liaise with a Medical Officer regarding further shortening.

- Some drains (e.g. Bore Drains) may be cut to fit a wound management bag instead of being shortened; clarify which type of drain is insitu.³

3.2 Nursing Management

- Once a drain has commenced being shortened, be aware the drain may slip out. Inform medical team if this occurs so the drain site can be managed appropriately.

3.3 Procedural Information

Requirements:

- Wet strength bag
- Dressing pack
- Sterile normal saline 0.9%
- Sterile scissors
- Sterile safety pin
- Sterile stitch cutter (if required)
- Gauze (if required)
- Adhesive dressing (if required)
- Wound management bag (if required)
- Skin Barrier wipe (e.g., Cavilon No-Sting Wipe)
- Personal Protective Equipment (PPE) including sterile gloves

Procedure

1. Perform hand hygiene.
2. Clean the trolley with combined detergent disinfectant wipe. Allow to dry.
3. Gather equipment for the procedure.
4. Perform hand hygiene and prepare the aseptic field.
5. Open dressing pack using corners. Open the remaining equipment onto the aseptic field using a non-touch technique.
6. Perform hand hygiene.
7. Using forceps, cleanse the insertion site with sterile normal saline 0.9%.¹⁻⁶
8. Remove suture if present.
9. Withdraw drain by the length documented by the Medical Officer.
10. **Don sterile gloves** and insert sterile safety pin through the drain proximal to the skin, to prevent the drain retracting back into the wound.
11. Trim the excess external length.
12. Protect the surrounding skin with skin barrier wipe.
13. Apply wound management bag/dressing over the drain.
14. Dispose of waste.
15. Remove PPE and perform hand hygiene.
16. Clean the trolley with combined detergent disinfectant wipe. Allow to dry.
17. Perform hand hygiene.
18. Document amount (length, in centimetres) drain is withdrawn in inpatient record.

4.0 Emptying Drain

4.1 Safety Information

Identify the type of drain insitu, as this will influence whether the drain may be emptied.

4.2 Nursing Management

- If the volume of the drainage is more than the previous day's drainage, liaise with Medical Officer.

- If the drain is attached to a drainage bag, ensure the drainage bag is below the insertion site, DO NOT secure to the bed; use a bedside bag holder to prevent contamination. The drainage bag must not be placed on the floor.

4.3 Procedural Information

Requirements

- Clean measuring container (e.g., urinal)
- Large 70% alcohol-impregnated wipe
- Personal Protective Equipment (PPE)

Procedure

1. Perform hand hygiene.
2. Gather equipment for the procedure.
3. Perform hand hygiene and don PPE.
4. Clean the drainage bag outlet/port with a large 70% alcohol-impregnated wipe.
5. Open the drainage bag outlet/wound management bag and empty drainage into a clean measuring container.
6. Check patency of drain refer to CNS or RMO if a clot is occluding tubing.
7. Close drainage outlet / re-secure wound management bag clip.
8. Dispose of drainage contents appropriately.
9. Remove PPE and perform hand hygiene.
10. Document amount Fluid Balance Chart. Description of drainage (as required) should be documented in the inpatient notes.

5 Removal of Drain

5.1 Medical Requirements

The removal of the drain is at the direction of the medical team and must be ordered and documented in the patient notes.

5.2 Safety Information

If the volume of the drainage is more than the previous day's drainage, liaise with Medical Officer regarding the removal of the drain.

5.3 Nursing Management

- For removal of Intercostal Drainage Tube, refer to NPG Chest Drain Management.
- For removal of PTC Catheter, refer to NPG - Percutaneous Biliary Drainage Catheter.
- Ensure suction has been turned off before removal.
- Give analgesia 30 minutes before removal as required.
- Rotating drain may assist in the ease of removal.⁵

5.4 Procedural Information

Requirements

- Wet strength bag
- Dressing pack
- Sterile normal saline 0.9%
- Sterile stitch cutter if required
- Personal Protective Equipment (PPE) including eye protection
- Appropriate dressing
- Wound management bag (if required)

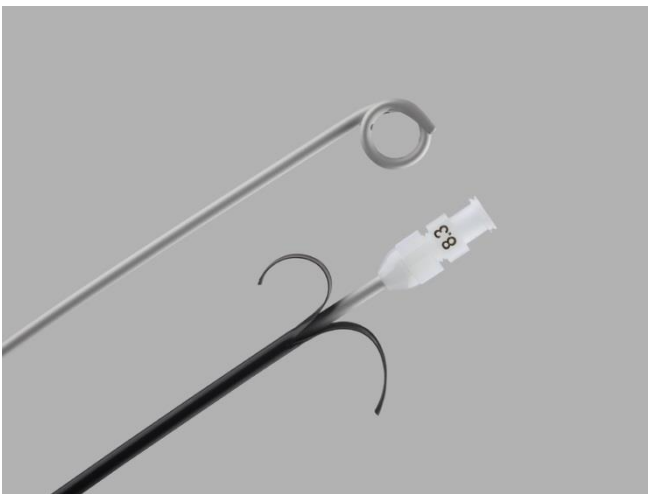
Procedure

1. Ensure suction is turned off for a minimum of five minutes before removing drain then apply slide clamp on drainage tubing.⁷

2. Perform hand hygiene.
3. Clean the trolley with combined detergent disinfectant wipe. Allow to dry.
4. Gather equipment for the procedure.
5. Perform hand hygiene and prepare the aseptic field.
6. Open dressing pack using corners. Open the remaining equipment onto the aseptic field using a non-touch technique.
7. Repeat hand hygiene.
8. Using forceps, cleanse the insertion site with normal saline 0.9%.¹⁻⁶
9. Remove suture if present.
10. Don PPE (non-sterile gloves) and remove the drain in a continuous motion by exerting gentle traction on the tube, while supporting skin surrounding the drain.⁵
11. Apply dressing/wound management bag to drain site.
12. Dispose of waste.
13. Remove PPE and perform hand hygiene.
14. Clean the trolley with combined detergent disinfectant wipe. Allow to dry.
15. Perform hand hygiene.
16. Document removal of the drain in the patient notes and record amount of drainage on the Fluid Balance Chart (MR 904).

6.0 Management of Abdominal Pigtail Drain

Pigtail Catheters are soft lumen tubes with a coil in the shape of a pigtail. They are used for percutaneous external drainage of a single cavity to remove pus, blood, or other body fluid, preventing it from accumulating in the body. Pigtail catheters can be locking or non-locking. The tips of pigtail drains are located on the inside of the pigtail curve to prevent irritation of the surrounding tissues. Depth markers aid in accurate placement, and also act as a reference for correct positioning.⁸ⁱ



6.1 Medical Requirements

Abdominal pigtail catheter insertion requires written consent as per the Western Australian Department of Health Consent to Treatment Policy.

Abdominal Pigtail Catheters are inserted under radiological guidance by a Medical Officer.

Flushing of a catheter is only to be performed on request from the Medical Officer and is to be prescribed on the WA Hospital Medication Chart. Instructions must include the volume of the flush and if the flush is to be aspirated or left in situ to drain.

Removal of the pigtail catheter is at the direction of the medical team and must be documented in the patient notes.

6.2 Safety Information

Observe for procedure related complications including bleeding, pain, sepsis, bowel perforation. Report any signs of frank blood in drainage fluid to medical officer.

Pigtail catheters used for percutaneous external drainage of a single abdominal cavity have either a non-locking or locking mechanism e.g. Merit medical resolve™ or Cook mac lock™.

Locked Pigtail catheters must be unlocked before removal. The mechanism for unlocking is dependent on catheter type (as per manufacturer's instructions: see below). Do not cut a pigtail catheter unless specified as suitable to do so by the manufacturer.

If the pigtail drain does not unlock it is responsibility of the medical officer to remove

When flushing pigtail catheter if any excess pressure is required, or if the patient experiences pain, stop the procedure and notify the Medical Officer.

All injectable medicines drawn up in a syringe should be labelled immediately. Labelling is not required when the preparation and bolus administration of a single medicine is one continuous process, and where the syringe does not leave the hands of the person who prepared it, and that same person administers the medicine.

Post removal, check that the catheter tip is intact, and document in the inpatient's notes/ medical record.

6.3 Nursing Management

Pre-Insertion

- A Radiology checklist must be completed for all patients having an invasive procedure.
- Refer to Nursing Practice Guideline 26: Radiological Examinations for pre-procedure preparation and care.

Post Insertion

- Refer to post procedure instructions.
- The catheter should be connected to a 3 way tap then to connecting tubing (CT) which has a 2 way on / off tap. The end of the CT may be spigotted or connected to a drainage bag. Post flushing procedure the pigtail catheter must have a drainage bag attached. If connected to a drainage bag check connection, patency and that there are no kinks in the tubing.
- If a 3-way tap is not connected – attach one between the catheter and the vinyl connecting tubing (Rocket drain connection set 30cm Rocket medical LOT 488492) using an aseptic non-touch technique. Place needleless bung on the middle port of 3-way tap. Place Mepilex™ Border or Sponge foam under 3-way Tap as a barrier to skin for comfort to relieve pressure.
- The catheter will be secured at the insertion site with a fixation device e.g., Statlock™ or Griplock™. A Statlock™ must be the correct size for the gauge of catheter; a Griplock™ is one size for all catheters.
- A Statlock™ will provide a more secure fixation than a Griplock™ and should be considered if accidental removal or longer-term use is anticipated e.g. restless patient, patients being discharged home with a pigtail catheter in situ. Consult with medical officer for consideration of suture for anchoring abdominal pigtail catheter (suture to be inserted by medical officer)

- A Statlock can be uncomfortable for the patient to lie on and may cause pressure on the skin, particularly if multiple catheters are in situ. Inspect the skin frequently for injury and educate the patient to avoid lying on the drain(s)/drainage bag.
- Anchor the drainage bag tubing to the skin with tape (Fixomul TM) or a catheter fixation device (e.g. Flexi Trak) to prevent excessive traction.
- If free drainage is ordered ensure both the 3-way tap and 2-way on/off tap of Connecting tubing is in the open position and drainage bag with a catheter tip connection is attached.
- Remove and replace primary fixation device Statlock™ or Grip-Lok™ every 7 days and PRN if soiled or lifting off the skin.
- Change secondary occlusive dressing as required. A small absorbent pad may be required under the occlusive dressing, if there is any fluid leak around the catheter insertion site, change dressing PRN to protect surrounding skin.
- Empty and record the volume of drainage each shift: record on fluid balance chart Total 24-hour drainage volume must be recorded on Fluid Balance Summary. Drainage can vary in colour and consistency; it may be golden yellow, brown, purulent, haemoserous, or occasionally clear to white fluid.
- Observe the insertion site for leakage, inflammation or discharge and report to Medical Officer. Report any pain at insertion site.
- When planning for patients to be discharged home with abdominal pigtail catheter in situ, ensure education² and referral to HomeLink or Silver Chain for ongoing care and monitoring on discharge.
 - Referral is to include a medical summary of accepted parameters for drainage volumes and flushing prescription to prescribe on the WA Hospital Medication Chart.
 - Instructions must include the volume of the flush and if the flush is to be aspirated or left in situ to drain.

6.4 Procedural Information

Flushing Abdominal Pigtail Catheter

Requirements

- Personal Protective Equipment (PPE) Non-sterile gloves
- Non-sterile kidney dish/blue tray
- Drawing up needle
- 10ml Syringe
- 0.9% Sodium Chloride
- Large 70% alcohol-impregnated wipe
- Absorbent sheet
- Catheter Tip Drainage Bag
- Rocket Drain Connection

Procedure

1. Perform hand hygiene
2. Assemble equipment in non-sterile kidney dish/blue tray and check patient identity against prescription order in WA Hospital Medication Chart
3. Position patient for ease of access to the catheter placing an absorbent sheet under the catheter.
4. Perform hand hygiene and don non-sterile gloves.
5. Draw up prescribed flush volume in 10ml luer lock syringe using drawing up needle.

6. Protect key parts using a non-touch technique. Scrub the needleless bung on the outlet port of the 3-way tap with a large 70% alcohol-impregnated wipe for 30 seconds. Allow to air dry.
7. Turn the 3-way tap off to the drainage bag and connect 10ml luer lock syringe with Sodium Chloride 0.9% flush to needleless bung.
8. Aspirate gently to commence flow and gently inject prescribed volume of Sodium Chloride 0.9% into catheter, do not use force.
9. Either aspirate gently to withdraw the same amount of fluid (if unable to withdraw fluid inform medical officer) **OR** leave fluid in if ordered.
10. Turn the 3-way tap - off to the needleless bung and disconnect the syringe.
11. Ensure both 3-way tap, and 2-way on/off tap of CT (rocket drain connection tubing) are left in open position to the drainage bag.
12. Check the system for patency and kinking.
13. Dispose of waste.
14. Remove PPE and perform hand hygiene.
17. Record completion of Sodium Chloride 0.9% flush on WA Hospital Medication Chart; if any difficulties are encountered inform the medical officer, shift coordinator and record in inpatient record.

Removal of Abdominal Pigtail Catheter

Consider analgesia 30 minutes before removal.

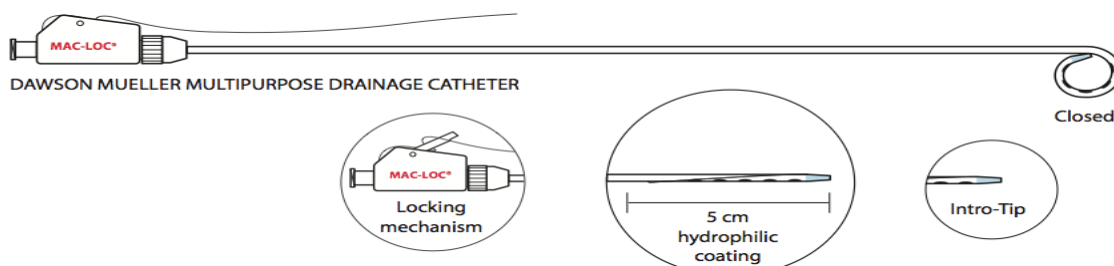
Requirements

- Wet strength bag
- Dressing pack
- Sterile normal saline 0.9%
- Sterile stitch cutter if required
- Sterile scissors
- Sterile dressing (e.g. Mepilex™ border) ¹⁻⁶

Personal Protective Equipment (PPE) Procedure

1. Perform hand hygiene.
2. Clean the trolley with combined detergent disinfectant wipe. Allow to dry.
3. Gather equipment for the procedure.
4. Perform hand hygiene and prepare the aseptic field.
5. Open procedure pack using corners. Open the remaining equipment onto the aseptic field using a non-touch technique.
6. Repeat hand hygiene and don PPE (non-sterile gloves).
7. Remove the stabilisation device.
8. Remove PPE and perform hand hygiene.
9. Using forceps, cleanse the insertion site with normal saline 0.9%.¹⁻⁶
10. Remove suture if present.
11. Unlock catheter if required (as per manufacturer's instructions)
 - a. Cook Medical: Mac-Loc™ locking mechanism

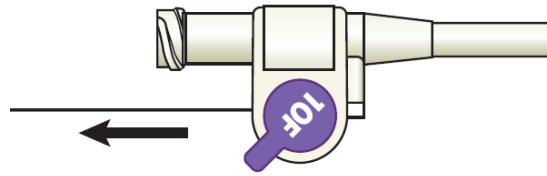
Locked position



To unlock:

- a) While stabilising the Mac-Loc catheter hub assembly with one hand, position a small blunt object (approximately the size and shape of a ballpoint pen or small forceps) into the Mac-Loc release notch.
- b) Pry forward until the locking cam lever is free.

A. Merit Medical Resolve™ Drainage catheter unlocking instructions



To unlock:

- a) Using the repositioning tool, align the opening of the round section of the tool in line with the handle of the suture locking mechanism.
- b) Bring the flat back of the repositioning tool around the catheter hub and gently squeeze together.
- c) Remove the repositioning tool and rotate the suture locking mechanism to the most proximal position. The hub is now unlocked.

1. Don PPE (non-sterile gloves).
2. Remove the drain in a continuous motion by exerting gentle traction on the tube, while supporting skin surrounding the drain.
3. Apply dressing to drain site.
4. Dispose of waste.
5. Remove PPE and perform hand hygiene.
6. Clean the trolley with combined detergent disinfectant wipe. Allow to dry.
7. Perform hand hygiene.
8. Document removal of the drain in the patient notes and record amount of drainage on Fluid Balance Chart⁶

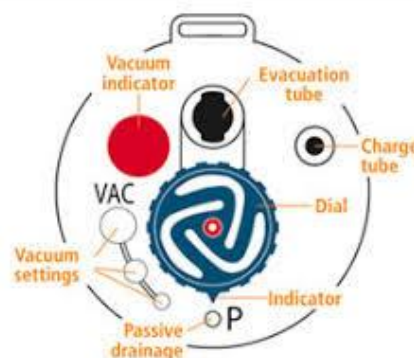
7.0 Suction Drainage

7.1 Safety Information

Drain Types:

- SCGH: Varivac™

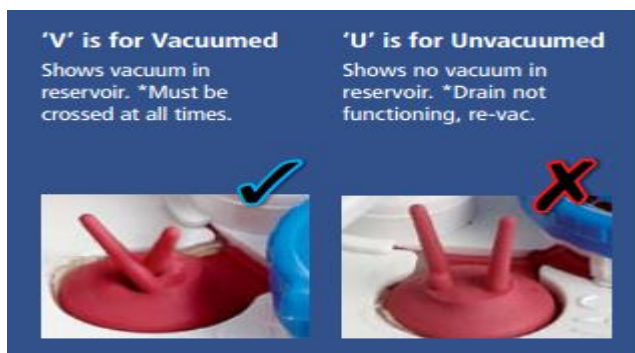
Varivac™ ensure suction adjustment dial is turned to "Passive (P - as shown below) and both slide clamps applied before changing drainage bottles.⁷



Suction drainage reservoir is a closed drainage system, and the reservoir may be placed on the floor.

7.2 Nursing Management

- Varivac™ wound drainage system has a blue suction adjustment dial, which should be set according to medical orders. Settings should be documented in the postoperative patient record. Suction may be adjusted up or down between the 4 settings if medical orders change.⁷
- The red vacuum indicator on the top of the bottle must be crossed making a "V" at all times indicating vacuum in the bottle. Uncrossed making a "U" indicates the system is unvacuumed and therefore not functioning. If suction is lost, discuss with surgical team if gravity (passive) drainage is sufficient, or if suction must be maintained change to a new bottle ⁷.
- As a minimum, the Varivac™ bottle must be marked daily at the fluid drainage level with a line, date and time at 2400hrs (and as ordered by the medical team). The result should be recorded on Fluid Balance Chart & Fluid Balance Summary. The need for more frequent assessment of drainage should be determined by the patient's clinical condition. Mark the Varivac™ bottle with a line and the time each time drainage is recorded on Fluid Balance Chart.



Passive P		0kPa
Low	I	5kPa (37mmHg)
Medium	II	10kPa (75mmHg)
High	III	15kPa (113mmHg)

7.3 Procedural Information

To Change Bottle

Requirements

- Varivac™ Bottle
- Large 70% alcohol-impregnated wipe
- Personal Protective Equipment (PPE)

Procedure

1. Perform hand hygiene and don PPE.
2. Turn blue suction adjustment dial to gravity then clamp tubing on both sides of luer lock.⁷
3. Open the packaging of new Varivac™ bottle and check slide clamp is applied, and the red vacuum indicator is crossed (V) ⁷
4. Disconnect the full Varivac™ bottle at the luer lock connector and clean the proximal end of the drain using the large 70% alcohol-impregnated wipe, allow to air dry.
5. Attach new Varivac™ bottle ensuring luer lock is secured. ⁷
6. Regulate suction level to the prescribed setting. ⁷
7. Release both tubing clamps. Check red vacuum indicator remains crossed (V). ⁷
8. Place the new bottle below the level of the insertion site.
9. Dispose of old drainage bottle in clinical waste.
10. Remove PPE and perform hand hygiene
11. Record drainage on Fluid Balance Chart.

Varivac™ Pathology Sampling

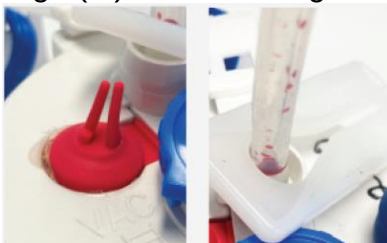
Specimens from varivac™ drainage canisters are often requested to be sent for testing in PathWest. To obtain a sample of drainage fluid from the canister, first change bottle as per above directions.

Requirements

- 50mL catheter tip syringe
- Yellow topped specimen container
- Personal Protective Equipment (PPE) including eye protection.
- Pathology request form
- Old varivac canister

Procedure

1. Perform hand hygiene and don PPE
2. Establish the old varivac canister is without vacuum (red indicators uncrossed showing 'U' for unvacuumed). If vacuum is present, release by turning blue dial to High (III) and releasing the white slide clamp on the silicone tubing.⁹



3. Remove the blue dust cap from the re-vacuuming port. Retain to reseal canister.



- 4.
5. Connect the catheter tip syringe by pushing it firmly into the re-vacuuming port.



Invert the canister until drain fluid covers the blue valve inside the canister and withdraw to extract the drainage sample.

6. Once completed, turn canister upright and put the white slide clamp back on the silicone tube.
7. Disconnect the syringe and replace blue dust cap. Slowly transfer specimen to yellow topped container.
8. Dispose of old Varivac canister and syringe into the clinical waste.
9. Remove PPE and perform hand hygiene.
10. Label sample and place in specimen bag with completed pathology form.

Varivac™ Re-vacuuming

Requirements

- 50mL catheter tip syringe (2-3 syringes required as each syringe is single use)

- PPE, including eye protection
- Kidney dish/blue tray
- Large 70% alcohol-impregnated wipe

Note:

- A new syringe must be used for a single withdraw of air from Varivac.
- Re-vacuuming should not occur when the bottle is full & requires changing

Procedure

1. Perform hand hygiene
2. Don PPE
3. Remove the blue dust cap from the re-vacuuming port and place in clean kidney dish/blue tray.
4. Clean the port using the large 70% alcohol-impregnated wipe and allow to air dry.
5. Using an aseptic non touch technique, attach the syringe and withdraw the plunger to expel air from Varivac.
6. Discard syringe and insert new syringe if required
7. Replace blue dust cap
8. Doff PPE and perform hand hygiene

Related Documents**Legislation**

- Work Health and Safety Act 2020
- Medicines and Poisons Act 2014 and the Medicines and Poisons Regulations 2016.
- Nursing and Midwifery Board of Australia (NMBA) Code of Professional Conduct for Nurses 2018

Policy

- Western Australian Department of Health Consent to Treatment Policy,
- NMHS Consent to Treatment Policy
- NMHS Clinical Documentation Policy
- SCGOPHCG Infection Prevention and Management Manual- Aseptic Technique Policy
- NMHS Policy Patient Identification & Procedure Matching
- SCGOPHCG Hospital Policy Patient Identification
- SCGOPHCG Hospital Policy Clinical Handover- Communicating for Safety.

Guidelines

- NPG Chest Drain Management- for information related to removal of Pleural Pigtail Catheter.
- SCGH Nursing Practice Guideline Percutaneous Nephrostomy Catheter.
- SCGH Nursing Practice Guideline Percutaneous Biliary Drainage Catheter.
- SCGH Nursing Practice Guideline Radiological Examinations and Neurointerventional Procedures.

References

1. Yimei Li. Evidence summary: Wound drainage site [Evidence Summary on the Internet]. Adelaide: Joanna Briggs Institute; 2013 [updated 2017 January 13; cited 2017 Feb 10]. Available from: JBI Connect + level of evidence : guideline
2. Joanna Briggs Institute. Wound drain: Dressing after surgery [Evidence Based Recommended Practice on the Internet]. Adelaide: Joanna Briggs Institute; 2013 [updated 2017 January 25; cited 2017 Feb 10]. Available from: JBI Connect + level of evidence : guideline
3. Joanna Briggs Institute. Closed wound suction drainage: Shortening [Evidence Based Recommended Practice on the Internet]. Adelaide: Joanna Briggs Institute; 2013 [updated 2017 January 25; cited 2017 Feb 10]. Available from: JBI Connect + level of evidence : guideline
4. Joanna Briggs Institute. Closed wound suction drainage: Emptying [Evidence Based Recommended Practice on the Internet]. Adelaide: Joanna Briggs Institute; 2013 [updated 2017 January 25; cited 2017 Feb 10]. Available from: JBI Connect + level of evidence : guideline

5. Joanna Briggs Institute. Closed wound suction drainage: Removal [Evidence Based Recommended Practice on the Internet]. Adelaide: Joanna Briggs Institute; 2013 [updated 2017 January 25; cited 2017 Feb 10]. Available from: JBI Connect + level of evidence : guideline
6. Joanna Briggs Institute. Closed wound suction drainage: Maintenance [Evidence Based Recommended Practice on the Internet]. Adelaide: Joanna Briggs Institute; 2013 [updated 2017 January 25; cited 2017 Feb 10]. Available from: JBI Connect + level of evidence : guideline
7. VariVac™. Product information: VariVac. [Internet]. Balcatta, WA (Australia): VariVac; 2012 [updated 2012: cited 2017 Feb 10] Available from: <http://www.varivac.com/canister.html>
8. Product information: Multipurpose drainage catheter: Instructions for use [Internet]: Cook Medical; 2015 [cited 2020 June 15]. Available from: www.cookmedical.com/data/IFU_PDF/T_MULTI_REV4.PDF (Product information).
9. VariVac™. Product Information: VariVac Pathology Sampling Guide [updated 2024.July 03: cited 2024 July]

Authorisation & Version Control

Policy Sponsor	Nursing Practice Guideline Subcommittee				
Policy Contact	Nursing Practice Guideline Subcommittee- Chair				
Date First Issued:	November 2001	Last Reviewed:	July 2024	Review Date:	January 2027
Version No.	9.1				
Clinical Reviewer:	C Liddell CNS SUU/ PAC S Chambers, A/Deputy Nurse Codirector – Surgical Services H Zagar CNS SCGH G61 J Connelly CNS OPH Ward 6 Update: July 2024- M Collins CNS G62				
Research Reviewer	C Liddell CNS Short Stay Unit /Preadmission				
Executive Sponsor	Deputy Nurse Co-Director Cancer Imaging and Clinical Specialties Service				
Endorsed by:	Nursing Practice Guideline Subcommittee			Date:	12/11/2024
Standards Applicable:	NSQHS Standards (Version 2): 				

For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

This document can be made available in alternative formats on request for a person with a disability.